

GARIL AI — Full Project Manual

GARIL AI (*Governed AI for Research, Instruction and Learning*) is a university-grade platform for academic research and paper writing. It combines agentic AI workflows, literature retrieval, citation-verified drafting, supervision tools, and institutional AI governance.

Built for universities, colleges, polytechnics, lecturers, and students (Nigeria and beyond). Product entity: TrustLed AI Ltd.

Related docs: [README.md](#) (quick start) · [PROJECT_WORKFLOW.md](#) (architecture deep-dive) · [deploy/README.md](#) (deployment hub)

Legacy naming: Packages, env vars, and tokens still use the older “feynman” brand (`feynman-web` , `feynman-backend` , `feynman_auth_token` , `FEYNMANN_*` , `NEXT_PUBLIC_FEYNMANN_BACKEND`). They refer to this same product.

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Part A — Product

1. Roles and access

Role	Who	Typical access
Student	Enrolled learners	Research Assistant (token-metered), notebook, supervised projects, assignments, feedback, notifications
Lecturer / researcher	Faculty	Same research tools + supervise projects, reviews, assignment briefs, students, analytics

University admin roles	Institutional governance staff	Governance console for their university (analytics, tokens, audit, alerts, incidents, users, reports, contributions, provenance, policies, privacy, retention) — filtered by feature permissions
Super admin (role === "admin")	Platform operators	Onboard universities, all users/admins, platform tokens, activities, research content, backups

Console role names (feature-permission matrix in `User` model): `governance_admin` , `faculty_admin` , `department_admin` , `compliance_officer` , `data_protection_officer` , `research_integrity_officer` , `auditor` , plus `viewer` .

Registration rules

- Users register against an **onboarded** (active) university. Platform admin accounts are exempt from that gate.
- Lecturers cannot register with free-email domains (institutional email required).
- Students use `/student/register` ; lecturers/researchers use `/register` .

2. Getting started

Login paths

Audience	Login	Register
Lecturer / researcher	<code>/login</code>	<code>/register</code>
Student	<code>/student/login</code>	<code>/student/register</code>
University admin	<code>/admin/login</code>	(provisioned by admin/super-admin)
Super admin	<code>/super-admin/login</code>	(bootstrap / env default admin)

Where you land after login

Role	Home
Student	<code>/student/dashboard</code>
Lecturer / researcher / others	<code>/dashboard</code>
University admin	<code>/admin</code>
Super admin	<code>/super-admin</code>

Public marketing landing: `/` . Health check page: `/healthz` .

```
flowchart TD
    Landing["/"] --> Auth{"Logged in?"}
    Auth -->|No| Login["Login or register"]
    Auth -->|Yes| Role{"Role"}
    Login --> Role
    Role -->|lecturer| Dash["/dashboard"]
    Role -->|student| SDash["/student/dashboard"]
    Role -->|admin roles| Admin["/admin"]
    Role -->|super admin| Super["/super-admin"]
```

3. Lecturer workspace

Sidebar and dashboard tools are defined in `lib/aula-nav.ts` .

Canonical research flow

flowchart TD
 Dash["/dashboard"] --> RA["/research"]
 RA --> Ideas["Generate ideas"]
 Ideas --> Outline["/research/outline"]
 Outline --> Gen["/research/generate"]
 Gen --> Paper["/research/paper"]
 Paper --> Saved["/research/saved"]

Step	Route	What you do
1. Ideas	/research or /research/[scope]	Pick discipline, topic, and research type (scope); generate idea options
2. Outline	/research/outline	Build a literature-backed Markdown outline
3. Configure	/research/generate	Citation style and paper options
4. Draft	/research/paper	Stream a full paper via the Research Assistant (chat-paper workflow)
5. Library	/research/saved	Open persisted papers
Assignment path	/research/assignment	Assignment-oriented research entry

Lecturer route map

Feature	Routes
Dashboard	/dashboard, /dashboard/research
Research Assistant	/research, /research/[scope], /research/outline, /research/generate, /research/paper, /research/assignment, /research/saved
Research Notebook	/research/notebook, /research/notebook/[projectId]
Supervision	/supervision, /supervision/projects, /supervision/projects/[projectId], .../pages/[pageId]
Reviews	/reviews, /reviews/[id]
Assignments	/assignments, /assignments/new, /assignments/[id], /assignments/[id]/edit
Students	/students, /students/[id]
Analytics / notifications	/analytics, /notifications
Alternate project editor	/projects, /projects/[projectId], .../pages/[pageId]
Legacy chat	/chat

Supervision Assistant

Use **Projects** (/supervision/projects) to supervise theses, dissertations, and student research folders: pages/chapters, TipTap editing, topic approval, AI chapter review, scoring, document import/export, and notifications.

Assignments publish briefs and collect submissions. **Reviews** is the inbox for submitted chapters and drafts. **Students** lists supervisees.

4. Student workspace

Nav: `lib/student-nav.ts` . Research mirrors the lecturer funnel under `/student/research/*` .

Feature	Routes
Dashboard	<code>/student/dashboard</code>
Research Assistant	<code>/student/research</code> , <code>/student/research/[scope]</code> , <code>/outline</code> , <code>/generate</code> , <code>/paper</code> , <code>/assignment</code> , <code>/saved</code>
Notebook	<code>/student/research/notebook</code> , <code>/student/research/notebook/[projectId]</code>
Student Assistant hub	<code>/student/assistant</code>
Projects	<code>/student/projects</code> , <code>/student/projects/new</code> , <code>/student/projects/[id]</code> , <code>.../pages/[pageId]</code>
Assignments	<code>/student/assignments</code> , <code>/student/assignments/[id]</code>
Feedback	<code>/student/feedback</code> , <code>/student/feedback/[projectId]/[itemKey]</code>
Notifications	<code>/student/notifications</code>

Student AI usage is **token-metered** (see [Token quotas](#)). When the allowance is exhausted, generation and chat runs that deduct tokens will fail until an admin raises the quota.

5. Research scopes

Research Type (scope) controls document structure, per-section in-text citation floors, word targets, and minimum distinct bank references. Profiles live in `lib/research-scope-profiles.ts` (mirrored on the backend).

Scope	Label	Min distinct cites	Word target (approx.)
assignment	Assignment	20	1,900–2,100
conference	Conference paper	20	3,000–4,000
journal	Journal / Research Paper	25	4,000–6,000
project_report	Project report	20	3,000–5,000
proposal	Research proposal	20	2,000–4,000
faculty	Faculty / grant	30	4,000–6,000
undergraduate	Undergraduate project	25	6,000–8,000
thesis	Thesis	40	8,000–10,000
dissertation	Dissertation	50	10,000–12,000

Example: **Assignment** uses Introduction → Literature Review → Critical Analysis → Conclusion (no Methods/Results). **Journal** uses IMRaD-style headings. Longer forms (thesis/dissertation) use chapter-style structures and higher citation floors.

The LLM is instructed to cite only papers from the retrieved **citation bank** for that run.

6. Admin and governance console

Login: `/admin/login` . Nav groups: `lib/admin-nav.ts` → `ADMIN_NAV_GROUPS` .

Area	Routes	Purpose
Overview	<code>/admin</code>	Governance hub: usage, alerts, adoption, health
Usage	<code>/admin/analytics</code> , <code>/admin/tokens</code>	Adoption and token consumption
Accountability	<code>/admin/audit</code> , <code>/admin/alerts</code> , <code>/admin/incidents</code> , <code>/admin/users</code>	Audit log, alerts, incidents, user lifecycle
Reporting	<code>/admin/reports</code>	Management / Senate / auditor reports
Research integrity	<code>/admin/contributions</code> , <code>/admin/provenance</code>	AI contribution statements and provenance (titles may be privacy-protected)
Controls	<code>/admin/policies</code> , <code>/admin/privacy</code> , <code>/admin/retention</code>	Policies, privacy rules, retention/deletion

Access is scoped to the admin's **university** and **feature permissions**. Governance oversight does not expose raw research materials unless policy allows it.

Stub / removed UI

These paths **redirect to** `/admin` . Backend models/services may still exist:

- `/admin/risks`
- `/admin/compliance`
- `/admin/inventory`
- `/admin/approvals`

7. Super-admin

Login: `/super-admin/login` . Home: `/super-admin` .

Route	Purpose
<code>/super-admin</code>	Platform overview
<code>/super-admin/universities</code>	Onboard / manage universities
<code>/super-admin/universities/detail</code>	University detail
<code>/super-admin/users</code>	Platform-wide users
<code>/super-admin/admins</code>	University / platform admins
<code>/super-admin/tokens</code>	Token defaults and usage
<code>/super-admin/activities</code>	Platform activity
<code>/super-admin/research</code>	Research content oversight

Super admins bypass per-feature checks and can offboard universities, reset passwords, and run backups (via admin APIs).

8. Token quotas

Defaults (`backend/src/constants/student-tokens.ts`):

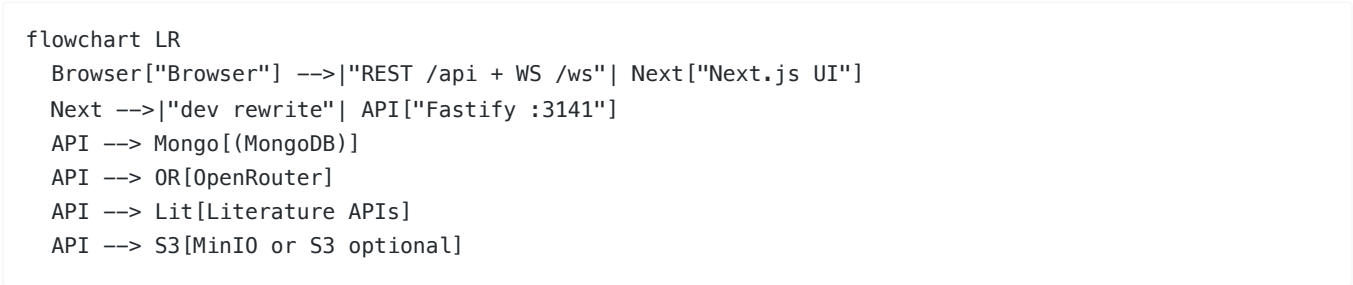
Role	Default allowance
Student	400,000 tokens
Lecturer / researcher	1,000,000 tokens

Resolution order: per-user override → university defaults → platform default.

Admins manage allowances under `/admin/tokens` (university) or `/super-admin/tokens` (platform). Remaining quota can appear in the UI and via WebSocket `student_token_quota` events during chat.

Part B — Technical / ops

9. Architecture



Layer	Stack	Default local URL
Frontend	Next.js 16, React 19, TypeScript, Tailwind 4	http://localhost:3000
Backend	Fastify 5, TypeScript, Node >=20.19 <26	http://127.0.0.1:3141
Database	MongoDB via Mongoose 8	<code>mongodb://127.0.0.1:27017/feynman</code>
AI	OpenRouter	—
Real-time	WebSocket <code>/ws</code>	Proxied via Next in dev
Editors / export	TipTap; jspdf, docx, pptxgenjs, mammoth, xlsx	—

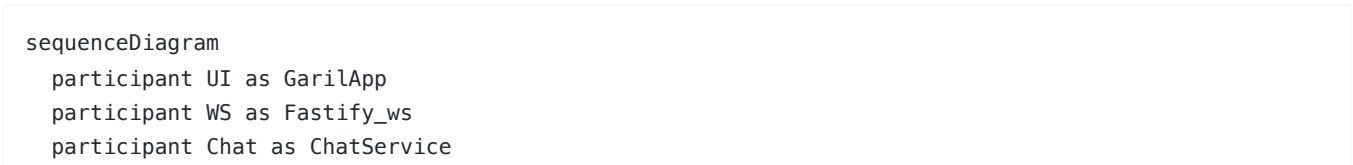
Frontend vs backend: Almost all business logic lives in Fastify. The Next app is the UI. In development, `next.config.ts` rewrites `/api/*` and `/ws` to `FEYNMAN_BACKEND_URL` (default `http://127.0.0.1:3141`). In production, set `NEXT_PUBLIC_FEYNMAN_BACKEND`, use a reverse proxy for same-origin, or serve static/standalone from the chosen deploy mode.

No separate workers: Research jobs run in-process (`research-jobs.service.ts`). No Redis/queue package.

Supabase: `@supabase/supabase-js` appears in `package.json` and optional `NEXT_PUBLIC_SUPABASE_*` in frontend `.env.example`, but **is not used** by application code. Primary database is MongoDB.

10. How the AI runtime works

Orchestration: `backend/src/services/chat.service.ts` over WebSocket `/ws`.



```

participant Lit as Paper_search
participant LLM as OpenRouter
participant DB as MongoDB

UI-->WS: reset workflow and topic
WS-->Chat: resetSession
Chat-->DB: Session plus system Message
WS-->UI: reset_complete
UI-->WS: prompt
WS-->Chat: sendMessage
Chat-->Lit: enrich first turn
Lit-->Chat: paper context
Chat-->LLM: stream
loop deltas
  LLM-->Chat: text_delta
  Chat-->UI: message_update
end
Chat-->DB: assistant Message optional SavedResearch
Chat-->UI: message_end token_usage agent_end

```

1. **Session** — `reset` loads a workflow prompt from `backend/prompts/*.md` , creates Mongo `Session` + system `Message` .
2. **Literature** (first user turn, research workflows) — builds a citation bank and injects abstracts/metadata into context.
3. **LLM stream** — OpenRouter; token deltas over WebSocket.
4. **Post-process** — format references (e.g. `chat-paper`), auto-save `SavedResearch` when long enough, deduct tokens.

Runtime truth: Prompt Markdown may describe tools (`web_search`) or subagents. The Fastify runtime does **not** run a general tool-calling agent loop. It uses **literature enrichment + LLM streaming**. Prompts still shape structure and quality.

Workflow prompts

Command	Description
<code>/chat-paper</code>	Full academic paper with inline citations (primary UI path)
<code>/deepresearch</code>	Multi-step investigation protocol (plan → gather → draft → cite → review → deliver)
<code>/lit</code>	Literature review grounded in retrieved papers
<code>/draft</code>	Structured section drafting
<code>/compare</code>	Side-by-side study comparison
<code>/summarize</code>	Concise summary
<code>/audit</code>	Citation / integrity audit
<code>/autoresearch</code>	Autonomous experiment-style loop
<code>/watch</code>	Monitor a topic
<code>/log</code>	Session journaling (skips literature inject)
<code>/replicate</code>	Replication planning
<code>/recipe</code>	Methodology builder
<code>/jobs</code>	Background task inspection (skips literature inject)

Other AI paths

Path	Service
Idea generation	research-ideas.service.ts
Outlines	outline.service.ts
Async paper jobs	research-jobs.service.ts
Citation align / format	citation-align.service.ts, research paper format services
Optional HF embed / NLI	huggingface.service.ts
Portal chapter AI review / chat	portal-ai.service.ts, lib/portal-review/*

Models: `FEYNMAN_MODEL` (main), `FEYNMAN_FAST_MODEL` (ideas/light), `FEYNMAN_OUTLINE_MODEL` (outlines).

11. Local development

Prerequisites

- Node.js `>=20.19.0 <26` (`.nvmrc` pins 25 ; helper `backend/scripts/with-node.sh`)
- MongoDB local or remote

Install

```
npm install
cd backend && npm install
```

(The repo also has `pnpm-lock.yaml` ; documented path uses **npm**.)

Environment

```
cp deploy/backend/.env.example backend/.env
```

Set at least `OPENROUTER_API_KEY` and a strong `AUTH_SECRET` for anything beyond casual local use. See [Environment reference](#).

Run

```
# Terminal 1 - UI
npm run dev

# Terminal 2 - API
npm run dev:backend
```

Service	URL
Frontend	http://localhost:3000
Backend	http://127.0.0.1:3141
Health	GET http://127.0.0.1:3141/api/health

Useful scripts

Script	Purpose
npm run build / build:all	Next + backend tsc
npm run start	node backend/dist/index.js
npm run lint	Next lint
npm run test:pdf	PDF smoke runner
npm run deploy:build*	Production packaging

Helpers: `dev-backend.sh` , `start-backend.sh` .

Smoke checklist

- 1. GET `/api/health` returns OK
- 2. Register/login as lecturer or student against an onboarded university (or use default admin)
- 3. Generate research ideas → outline
- 4. Open `/research/paper` and confirm WebSocket streaming
- 5. Admin: open `/admin` after admin login

12. Environment reference

Canonical accessors: `backend/src/config/env.ts`

Templates: `deploy/backend/.env.example` , `deploy/frontend/.env.example` , `deploy/minio/.env.example`

Never commit real secrets (`.env` / `backend/.env` are gitignored).

Core (backend)

Variable	Required	Purpose
OPENROUTER_API_KEY	Yes	LLM calls
AUTH_SECRET	Yes in production	HMAC JWT signing
MONGODB_URI	Yes in prod/Docker	Mongo connection (local default <code>mongodb://127.0.0.1:27017/feynman</code>)
PORT	No	API port (default 3141)
CORS_ORIGIN	No	Comma-separated allowed origins

Models / workspace

Variable	Purpose
FEYNMAN_MODEL	Main chat model
FEYNMAN_FAST_MODEL	Ideas / light tasks
FEYNMAN_OUTLINE_MODEL	Outlines
FEYNMAN_WORKSPACE / FEYNMAN_REPO_ROOT	Path overrides
OPENAI_MODEL	Portal AI naming fallback

Literature / ML

Variable	Purpose
ALPHAXIV_*	AlphaXiv (enabled, key, base, MCP URL)
ARXIV_API	arXiv export URL
TAVILY_ENABLED, TAVILY_API_KEY	Web search fallback
OPENALEX_*	OpenAlex
PUBMED_* / NCBI_API_KEY	PubMed / NCBI
DOAJ_*	DOAJ
EUROPE_PMC_*	Europe PMC
HF_TOKEN / HUGGINGFACE_*	Optional embeddings / NLI
PAPER_LIBRARY_ENABLED, PAPER_LIBRARY_MIN_HITS	Mongo paper-library RAG

Admin bootstrap

Variable	Purpose
DEFAULT_ADMIN_EMAIL / PASSWORD / NAME	Seeded platform admin
DEFAULT_ADMIN_ENABLED	Set false to disable auto-bootstrap
SEED_STUDENT_PASSWORD	Demo seed scripts only

S3 / MinIO (backend)

S3_ENDPOINT , S3_REGION , S3_BUCKET , S3_ACCESS_KEY , S3_SECRET_KEY , S3_FORCE_PATH_STYLE , S3_PUBLIC_URL , S3_MAX_UPLOAD_BYTES , S3_MAX_INLINE_BYTES

See [deploy/minio/README.md](#).

Frontend / Next

Variable	Purpose
NEXT_PUBLIC_FEYNMAN_BACKEND	Browser API origin (empty = same-origin)
FEYNMAN_BACKEND_URL	Dev rewrite target (default http://127.0.0.1:3141)
GARIL_STATIC_EXPORT	0 → Next standalone ; unset in prod often means static export to out/
NEXT_PUBLIC_SUPABASE_*	Documented optional; unused in current app code

13. Auth and tenancy

1. Register/login → password hash (backend/src/lib/password.ts) → HMAC JWT (~7-day TTL) via backend/src/lib/auth-token.ts and AUTH_SECRET .
2. Frontend stores token as feynman_auth_token in localStorage (lib/auth.ts); hooks/useAuth.tsx refreshes via GET /api/auth/me .
3. HTTP: Authorization: Bearer <token> . WebSocket: token on the connection query string.
4. **University gate**: only active onboarded universities may register/login (except platform admin).
5. **Portal authz**: requirePortalActor / student vs supervisor (backend/src/lib/portal-auth.ts); tenant key = universityId .

- 6. **Admin authz:** `requireAdmin` / `requireSuperAdmin` / `requireAdminScope` / `requireFeatureAccess` (`backend/src/lib/require-admin.ts`).
- 7. **No Next.js middleware** for auth — protection is client redirects + API enforcement.

14. API and WebSocket reference

All HTTP APIs are Fastify: `backend/src/server.ts` + `backend/src/routes/portal.ts` .
Frontend portal client maps `/api/v1/*` → `/api/portal/*` (`lib/portal-api.ts`).

Auth and system

Method	Path	Purpose
GET	<code>/api/health</code>	Liveness (may include S3 status)
GET	<code>/api/auth/universities</code>	Onboarded universities for signup
GET	<code>/api/auth/universities/catalogue</code>	Country catalogue (Hipolabs)
POST	<code>/api/auth/register</code>	Lecturer register
POST	<code>/api/auth/register-student</code>	Student register
POST	<code>/api/auth/login</code>	Issue JWT
GET	<code>/api/auth/me</code>	Current user + quota
GET	<code>/api/workflows</code>	Prompt workflow list
GET	<code>/api/status</code>	Chat service status
GET	<code>/ws</code>	Agent WebSocket

Research

Family	Purpose
GET <code>/api/papers/search</code>	Literature search
POST <code>/api/research/outline</code>	Literature-grounded outline
POST <code>/api/research/ideas/generate</code>	Idea generation
<code>/api/research/jobs*</code>	Async paper jobs (start / active / get / cancel)
<code>/api/research/saved*</code>	Saved papers CRUD
<code>/api/research/ideas/saved*</code> , <code>/sessions*</code> , <code>/outlines/saved*</code>	Ideas, sessions, outlines
<code>/api/research/projects*</code> , <code>/project</code> , <code>/workspace</code> , <code>/activity</code>	Notebook projects
<code>/api/research/documents*</code> , <code>/references*</code> , <code>/datasets*</code> , <code>/questionnaires*</code>	Notebook assets (+ upload sessions)
POST <code>/api/research/source-context</code> , <code>/visualizations</code> , <code>/outputs/sync</code>	Context, plots, sync
<code>/api/session/reset</code> , <code>/api/chat/abort</code> , <code>/api/sessions/:id/messages</code>	Chat control

Portal (`/api/portal/*`)

Projects (CRUD, pages, review, AI summary/chat, topic submit/approve, attach brief, score, import/export, images), chapters (approve/reject/save-review/ai-reviewer/versions), assignment briefs + submissions, feedback, notifications, deadlines, analytics

overview, students/supervisors lists, supervisor reviews inbox.

Admin (/api/admin/*)

Universities, users, admins, tokens, sessions, research papers/uploads, backup, governance overview/analytics, policies (+ evaluate), audit (+ flag), reports, incidents, alerts, contributions, provenance, privacy, retention (+ deletion requests).

Dashboard / legacy

GET /api/dashboard/stats , /api/dashboard/sessions , /api/users (+ :id).

WebSocket protocol

Client → server

Message	Purpose
{ type: "reset", workflow?, topic?, message? }	New session / workflow
{ type: "prompt", message }	User turn
{ type: "abort" }	Cancel run

Server → client

Message	Purpose
connected	Status + workflow list
agent_event	agent_start, tool_execution_*, message_update, message_end, token_usage, agent_end
reset_complete / prompt_complete / aborted / error	Lifecycle
student_token_quota	Remaining tokens

Paper client sequence (hooks/useGarilSocket.ts): reset with workflow: "chat-paper" → on reset_complete , send prompt with the full paper prompt.

15. Data model overview

MongoDB via Mongoose (backend/src/db/connect.ts). Models under backend/src/db/models/ .

Group	Models
Identity / tenancy	User, University
Chat	Session, Message
Research outputs	SavedResearch, SavedResearchIdea, SavedResearchOutline, ResearchIdeaSession, ResearchJob, OutputArtifact
Notebook	ResearchProject, ResearchDocument, ResearchDataset, ResearchReference, ResearchQuestionnaire
Literature cache	PaperLibrary
Supervision portal	PortalProject, PortalChapter, PortalChapterVersion, PortalAiReview, PortalNotification, AssignmentBrief

Governance	AuditLog, GovernanceAlert, GovernanceIncident, GovernancePolicy, GovernanceReport, GovernanceRisk
Integrity / privacy	AiContributionStatement, ResearchProvenanceRecord, ResearchPrivacySetting
Compliance (backend; some UI stubbed)	RetentionPolicy (+ DeletionRequest), ComplianceControl, AiSystemInventory, ApprovalRequest

Tenancy: primarily `universityId` on users and portal/governance documents.

16. Literature and citation integrity

Retrieval order (library-first RAG)

Priority used when building banks (see `PROJECT_WORKFLOW.md` and literature services):

1. **Mongo paper library** (if enabled and enough hits; default min hits = 4)
2. **AlphaXiv** (and MCP when keyed)
3. **arXiv**
4. **OpenAlex, PubMed, DOAJ, Europe PMC** (as configured / interleaved)
5. **Tavily** — web fallback when scholarly sources under-fill

New hits are upserted into the paper library for reuse (deduped by arXiv ID / title where applicable).

Citation rules

- The model should cite only papers present in the **retrieved bank** for that run.
- Scope profiles set minimum distinct bank references and per-section cite floors.
- Post-processing aligns and formats references (`citation-align.service.ts` , paper format services).
- Outline generation fetches up to ~8 papers, injects abstracts, then appends a “Sources for further reading” section.

17. Deployment

Hub: [deploy/README.md](#). **Not Vercel** — Coolify / Docker / VPS / Nixpacks. **No in-repo CI**.

Modes

Mode	Entry	Notes
Coolify split (recommended)	deploy/backend/Dockerfile + deploy/frontend/Dockerfile	API :3141 /api/health; UI :80 /healthz
Docker Compose	docker-compose.yml	mongo + backend + frontend
All-in-one	Root Dockerfile or deploy/all-in-one/	Single Node process
MinIO	deploy/minio/	Optional S3-compatible storage

Detailed Coolify/VPS/PM2 steps: [deploy/backend/README.md](#), [deploy/frontend/README.md](#), [deploy/all-in-one/README.md](#).

Frontend: standalone vs static export

Mode	How	Use when
Standalone (Coolify Dockerfile)	GARIL_STATIC_EXPORT=0 → Next standalone (node server.js on port 80)	Current deploy/frontend/Dockerfile path; supports dynamic App Router routes

Static export	Leave GARIL_STATIC_EXPORT unset for production output: "export" → out/ + nginx	Older VPS/nginx out/ deploys (deploy/frontend/build.sh + nginx confs)
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Build arg / env: `NEXT_PUBLIC_FEYNMAN_BACKEND` = public API URL (or empty for same-origin proxy).

Compose quick start

```
cp deploy/backend/.env.example backend/.env
export NEXT_PUBLIC_FEYNMAN_BACKEND=http://localhost:3141
docker compose up -d --build
```

18. Database ops

- **No migration runner** (no Prisma/Drizzle). Schema = Mongoose models. Change models carefully in production.
- **Connect** (`backend/src/db/connect.ts`): retries in production; index repair (e.g. legacy unique `userId` on research projects; paper-library `syncIndexes`).
- **Default admin**: created on boot via `bootstrap-admin.service.ts` unless `DEFAULT_ADMIN_ENABLED=false` . One-shot script: `backend/scripts/ensure-default-admin.ts` .
- **Demo seeds** (not general fixtures): `seed-okes-survey.ts` , `seed-okes-supervision.ts` (`SEED_STUDENT_PASSWORD`).
- **Governance cleanup** on boot may remove old mock rows (`admin-governance-cleanup.service.ts`).

19. Testing and CI

What exists	Command / path
Lint	<code>npm run lint</code>
Backend typecheck	<code>cd backend && npm run typecheck</code>
PDF smoke	<code>npm run test:pdf → scripts/test-research-pdf-runner.ts</code>

Missing: Jest/Vitest/Playwright/Cypress; no `*.test.*` / `*.spec.*` ; no `.github/workflows CI`.

Release path today: Coolify git deploy, Docker build, or VPS `setup-vps.sh` / PM2 / systemd. Suggested future CI: lint + `tsc` + health probe.

Recommended manual smoke: [Local development](#) checklist plus portal project create and admin token page.

20. File map and glossary

Key paths

Concern	Path
Product docs	<code>README.md</code> , <code>PROJECT_WORKFLOW.md</code> , <code>docs/PROJECT_MANUAL.md</code>
Deploy hub	<code>deploy/README.md</code>
Next config	<code>next.config.ts</code>
Env accessors	<code>backend/src/config/env.ts</code>
Server / WS	<code>backend/src/index.ts</code> , <code>backend/src/server.ts</code>

Portal routes	backend/src/routes/portal.ts
Chat / LLM	backend/src/services/chat.service.ts, llm.service.ts
Workflows	backend/src/services/workflows.ts, backend/prompts/
Auth	backend/src/services/auth.service.ts, backend/src/lib/auth-token.ts, hooks/useAuth.tsx, lib/auth.ts
Models	backend/src/db/models/
Research UI	components/ResearchAssistant.tsx, components/GarilApp.tsx, hooks/useGarilSocket.ts
API base URL	lib/api.ts
Role routing	lib/dashboard-routes.ts
Scope profiles	lib/research-scope-profiles.ts
Admin / student nav	lib/admin-nav.ts, lib/aula-nav.ts, lib/student-nav.ts

Glossary

Term	Meaning
GARIL	Governed AI for Research, Instruction and Learning
Feynman	Legacy internal/package name for the same product
Workflow	Markdown system prompt in backend/prompts/ conditioning the chat session
Citation bank	Set of papers retrieved and injected before generation
Scope / Research Type	Deliverable profile (assignment, thesis, ...) controlling structure and cite floors
Portal	Supervision / project / chapter / assignment subsystem (/api/portal/*)
Paper library	Mongo cache of scholarly metadata for library-first RAG
Token quota	Usage allowance for student/lecturer AI calls

Empty placeholders

app/settings/ and app/review-desk/ exist as directories without pages yet.

End of manual. For shorter setup instructions see [README.md](#). For sequence-level research pipeline detail see [PROJECT_WORKFLOW.md](#).